Troubleshooting

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When troubleshooting, it is helpful to rule out some of the more obvious situations that may lead to a fault condition:

• Check that the main power switch is in the ON position.

• Check that power is supplied to the main switch.

• Check that the sensors heads/lenses are clean.

• Check for faulty or pinched wiring and damaged, broken or worn parts.

• Make sure the correct product is being supplied.

• Check for product jams or misaligned guides.

• Check for recent changes to the system or to the environment in which it exists.

The human/machine interface [HMI] provides valuable information on the operational status of the machine. The operator interface display aids in the troubleshooting process by providing fault messages. These fault messages have been established to minimize the time spent in locating or identifying various anticipated machine fault conditions. As all possible faults cannot be anticipated some machine fault conditions may not trigger a message. Using materials for which the machine was not designed may cause a machine fault which may or may not be indicated in the operator interface display. Please reference the operation section in this manual for information on the HMI display and fault messages.

If there is a fault condition, and no message is displayed, it is good practice to rely on standard troubleshooting procedures. Efficient methodical troubleshooting starts with a clear understanding of the expected behavior of the system and the symptoms being observed. From there the you can form hypotheses on potential causes, and devise a series of tests to eliminate these prospective causes. Two common strategies are to check for frequently encountered or easily tested conditions first (for example, the itemized checks above), and to "bisect" the system. This is the application of a binary search across the range of dependences. One of the core principles of troubleshooting is that reproducible problems can be reliably isolated and resolved. Often considerable effort and emphasis in troubleshooting is placed on reproducibility, on finding a procedure to reliably induce the symptom to occur. Once this is done then systematic strategies can be employed to isolate the cause or causes of a problem; and the resolution generally involves repairing or replacing those components which are at fault.

Check each component in the system one by one, substituting known good components for each potentially suspect one. Start from the simplest and most likely problems first. Where identification of a fault or correction thereof proves troublesome it may be necessary to call for a **BluePrint** Automation factory-trained service technician (reference the contact information provided in the Introduction section of this manual).

Troubleshooting Table

Problem	Cause	Solution
Machine Overall		
Machine won't start.	Machine in fault status.	Remedy the fault condition and clear the fault.
	Emergency stop pushbutton pressed.	Reset Emergency stop pushbutton.
	Safety guard open.	Close safety guard.
Circuit breaker tripped.	Overcurrent or defective breaker.	Check current. Check efficacy of breaker, replace if defective.
VFD fault.	Machine not ready.	Press the stop button. Press the reset button. Power off and on at the main disconnect switch.
	Drive defective.	Repair or replace.
HMI		
Blue screen	System error.	Power off and on at the main disconnect switch. Check connections.
Photoelectric Sensors		
The sensor's output will not change state when an object enters the operating zone.	Sensor dirty.	Clean sensor.
	Sensor misalignment.	Readjust sensor.
	Output stage faulty or the short-circuit protection has tripped.	Check for wiring faults (short-circuit).
	Wiring error.	When replacing sensors, verify that the wiring conforms to the wiring shown on the proximity sensor label or instruction sheet.
	Sensor defective.	Check efficacy of sensor; replace defective device.
False or erratic operation, with or without the presence of an object in the operating zone.	Electromagnetic interference.	Check for sources of electromagnetic fields.
	Effect of interference on the supply lines.	Where very long distances are involved, use suitable cable: screened and twisted pairs of the correct cross-sectional area. Position the cables as far away as possible from any sources of interference.
	Influence of high-temperature.	Eliminate sources of radiated heat.
	Excess vibration.	Eliminate source of vibration. Make sure mounting hardware is secure.

Belt Conveyors		
Excessive slack in conveyor belt.	Normal wear.	Expect rapid belt growth in first few weeks of operation. Adjust belt tension as needed.
Motor/reducer running hot or hard to start.	Drag on conveyor.	Check entire conveyor for cause.
	Lack of lubricant.	Check for leaks, refill oil.
	Frozen pulley or roller.	Inspect all pulleys and bearings, replace as needed.
	Overload.	Reduce cause.
	Electrical problem.	Check Wiring and check amperage, replace as needed.
Excessive noise motor/reducer.	Lack of lubricant.	Check for leaks, refill oil.
	Damaged gears.	Replace unit.
	Faulty bearing.	Replace bearing.
Excessive wear, conveying belt or pulleys.	Excessive belt tension.	Adjust belt tension.
	Pulleys misaligned.	Realign pulleys.
	Damaged pulley.	Replace pulley.
	Dirty belt.	Clean belt.
Excessive noise, conveying belt or pulleys.	Insufficient belt tension.	Tension belt.
	Pulleys misaligned.	Realign pulleys.
Pulsating belt.	Insufficient belt tension.	Retension belt.
	Overload.	Reduce cause.
Broken belt.	Frozen bearing or shaft.	Inspect for frozen bearing, replace repair belt.
	Worn or damaged belt.	Repair/replace as needed.
	Jam.	Clear Jam.
Pulley loose.	Loose set screw.	Realign pulley, tighten set screw.
	Missing, worn or damaged key.	Replace with new key.

Synchronous Belt Drives		
Broken belt: Straight cord line rupture.	Belt was crimped prior to fitment.	Make sure to observe proper belt handling instructions and never coil, crimp or twist timing belts as this damages their tensile cords.
Broken belt: Raffled cord line rupture.	Foreign object between the belt and pulley will break the tensile cords; or belt was twisted over 90° or turned inside out before fitment, or forced on a pulley e.g. using a sharp object such as a screwdriver.	Make sure to observe proper belt handling instructions and never coil, crimp or twist timing belts as this damages their tensile cords.
Tooth shear.	Insufficient tension.	Install at correct tension.
	Seizure of driven part.	Eliminate cause.
	Misalignment.	Correct alignment.
Tooth wear.	Incorrect tension.	Adjust belt tension.
	Worn pulleys.	Replace pulleys.
Hollowed teeth.	Extremely low tension.	Adjust belt tension.
	Loss of tension during running.	Ensure tensioner screws are tight.
Back cracks.	Exceedingly high or low temperature.	Control ambient temperature.
	Back idler is worn out.	Replace back idler.
Land wear.	Excessive tension.	Adjust belt tension.
	Rough pulley(s).	Replace pulleys.
Oil contamination.	Oil leak.	Replace faulty oil seals.
Edge wear.	Damaged flange(s)	Repair/replace as needed.
	Misalignment.	Correct alignment.
Belt noise.	Excessive or insufficient tension.	Adjust belt tension.
	Misalignment.	Correct alignment.
	Damaged flange(s)	Repair/replace as needed.

Asynchronous Gearmotors		
Unusual running noises or vibrations.	Oil level too low.	Check oil level, adjust as needed.
	Bearing damage.	Replace the bearing if damaged.
	Gearwheel damage.	Replace the gearwheel if damaged.
Oil escaping from the gear unit or motor.	Defective seal.	Replace defective seals.
Oil escaping from pressure vent.	Incorrect oil level.	Check oil level, adjust as needed.
	Incorrect oil type or oil contaminated.	Change the oil.
	Unfavourable operating conditions.	Control operating conditions or install optional oil expansion tank.
Gearhead becomes too hot.	Unfavourable operating conditions.	Control operating conditions.
	Gear unit damaged.	Replace or repair damaged units.
Shock or vibrations when switching on.	Defective motor coupling.	Replace elastomer gear rim.
	Loose gear unit mounting.	Tighten mounting hardware.
	Defective vibration isolator.	Replace defective vibration isolator.
Output shaft does not rotate although the motor is running.	Fracture in gear unit.	Replace or repair damaged units.
	Defective motor coupling.	Replace damaged coupling.
	Shrink disc slippage.	Replace or repair damaged units.